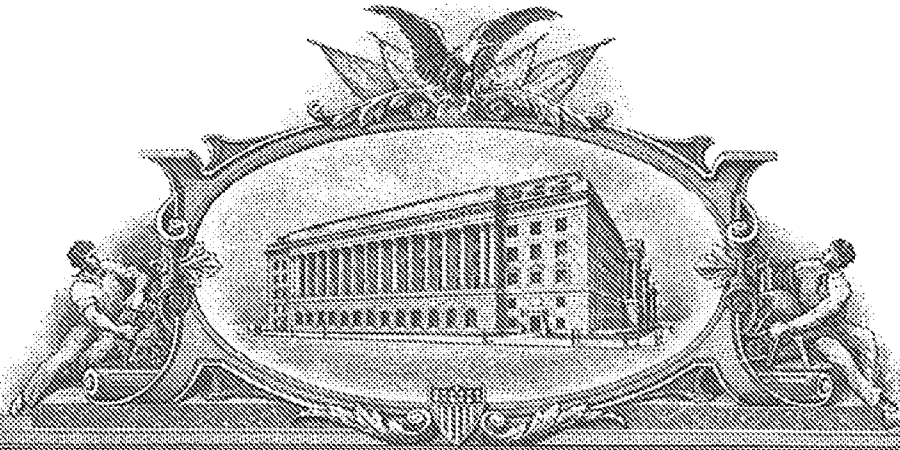


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Title of Invention:	METHOD AND SYSTEM FOR ARBITRARY WAVEFORM GENERATION USING A TRI-STATE TRANSMIT PULSER
First Named Inventor/Applicant Name:	John A. Flynn
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ABSTRACT OF THE DISCLOSURE

A method and system for generating arbitrary ultrasonic waveforms using a tri-state transmitter. Three variants of the device are described to provide functionality in three usage scenarios.

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		Application Number	
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	930125.406P1
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Attorney Docket Number	930125.406P1	Small Entity Status Claimed	☒
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Subject Matter	Utility		
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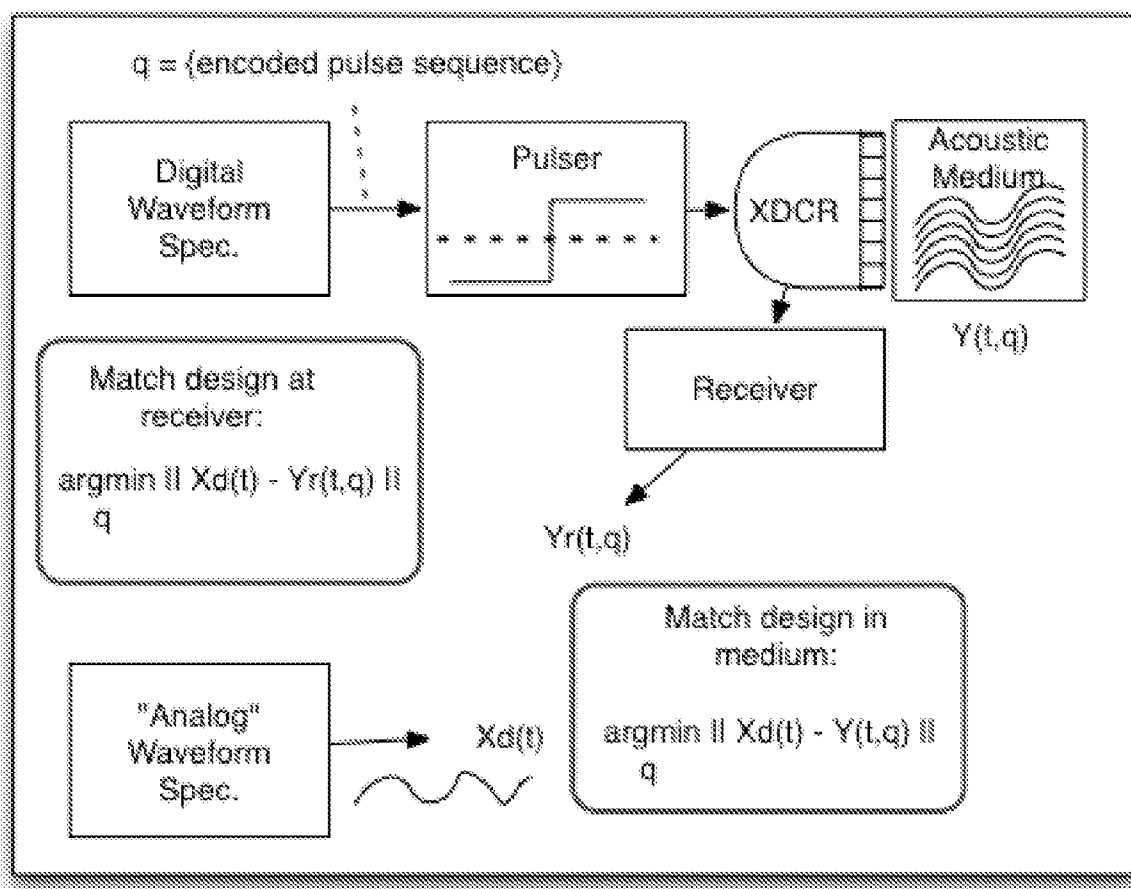


Figure 1

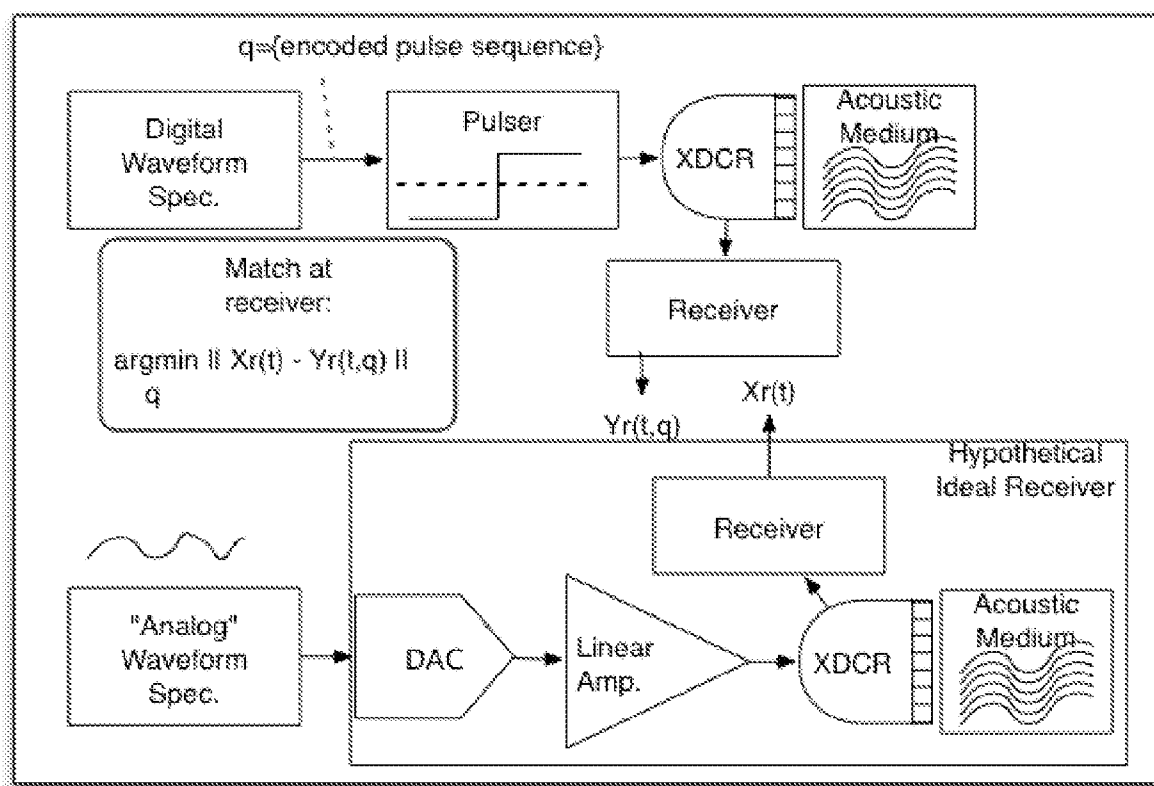


Figure 2

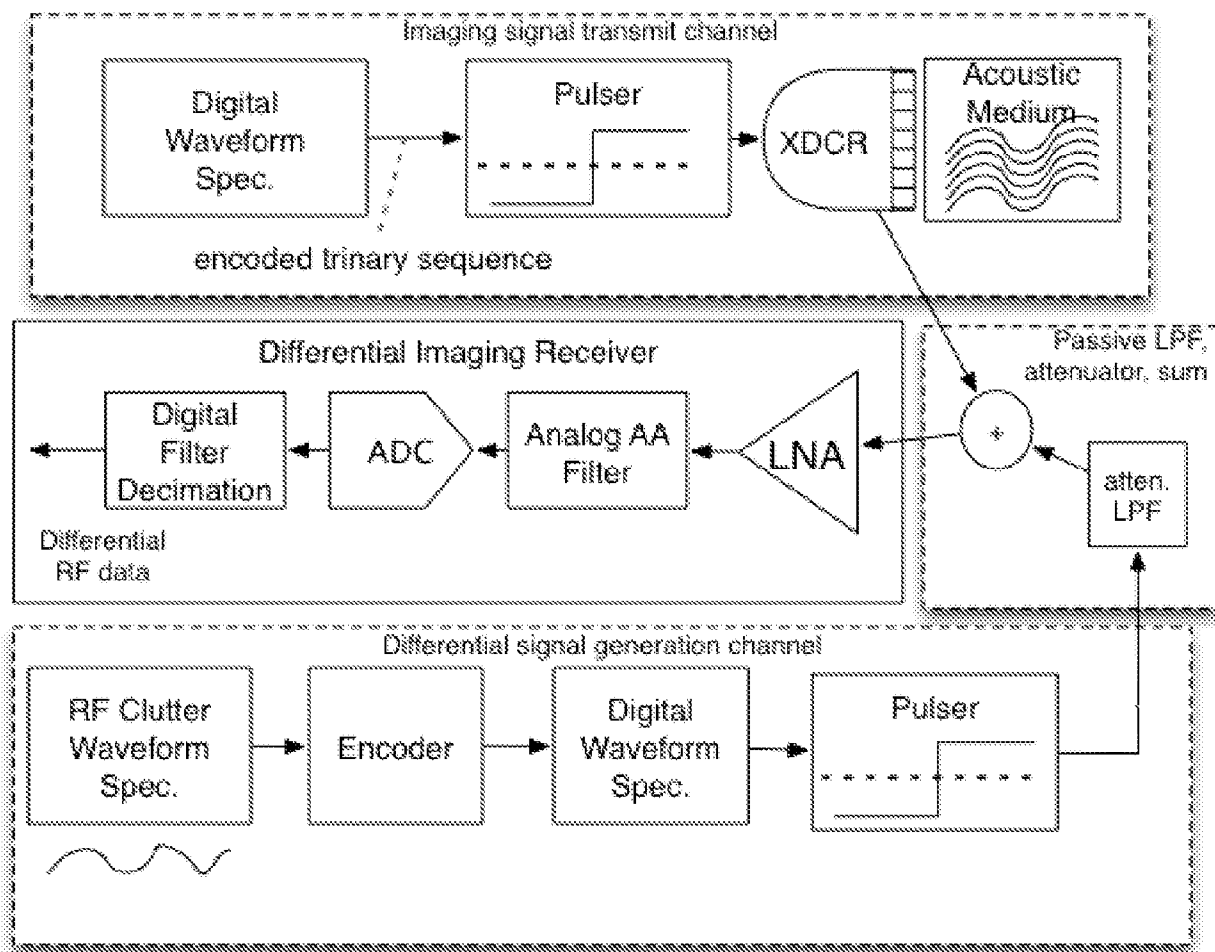


Figure 3

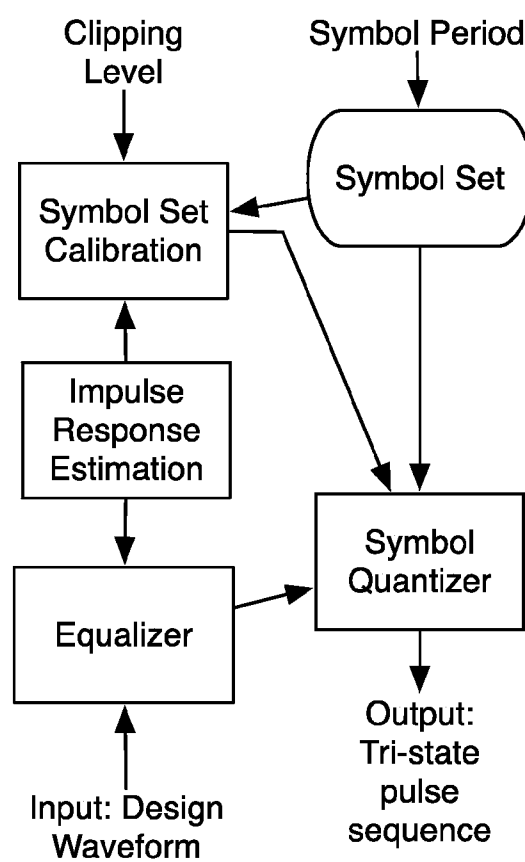


Figure 4

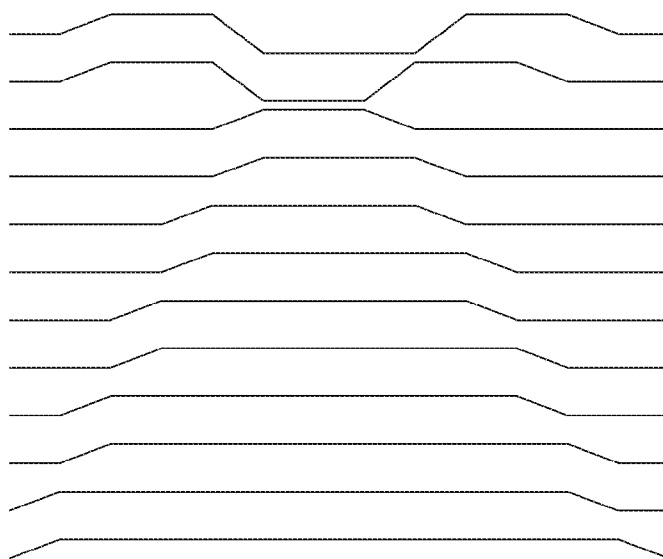


Figure 5

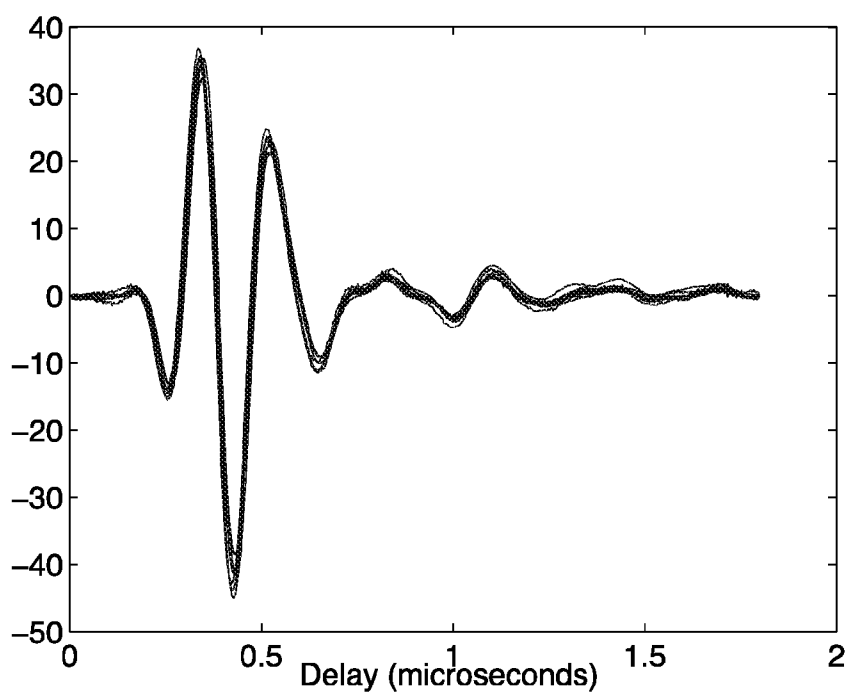


Figure 6

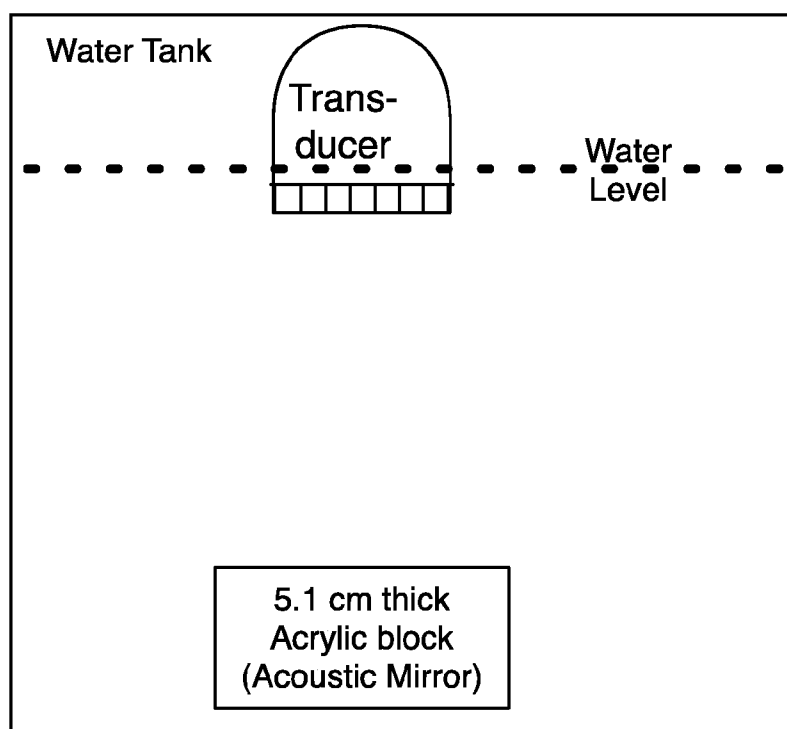


Figure 7

METHOD AND SYSTEM FOR ARBITRARY WAVEFORM GENERATION USING A TRI-STATE TRANSMIT PULSER

BACKGROUND

Technical Field

5 The present disclosure pertains to methods for encoding arbitrary waveforms into a sequence suitable for control of a tri-state RF ultrasonic transmitter, under various fidelity criteria, and to a related system.

BRIEF SUMMARY

10 The encoding methods and system disclosed herein require knowledge of the transducer element's impulse response (IR). An impulse response estimation method is disclosed and the results of the method are used to introduce encoding algorithms that optimize tri-state pulser sequences. The encoding algorithms are based on constrained deconvolution concepts from communications science known as "equalizers," combined with a hybrid pulse-width modulation (PWM) symbol modulation and quantization
15 scheme. Acoustic water-tank experiments with a Philips L7-4 transducer demonstrate fidelity of -21.7 dB normalized RMS error (NRMSE) in reproducing a windowed Linear Frequency Modulation (LFM) sweep signal.

20 The tri-state encoding concept disclosed herein has been implemented on the Vantage Ultrasound System manufactured by Verasonics, Inc. Redmond, WA (USA). In contrast to a digital-to-analog converter (DAC) driving linear RF amplifiers, the tri-state transmitter architecture of the present disclosure requires non-obvious selection of its pulse sequence to achieve fidelity to a continuous-valued design waveform. The process demonstrated here exploits the high transmitter clock frequency (with respect to transducer bandwidth) to achieve that goal.

25 In accordance with one aspect of the present disclosure, a method is provided that includes providing a binary or trinary sequence of symbolic values that are configured to

energize an ultrasonic transducer element with a corresponding sequence of positive, negative, or quiescent voltage levels at a corresponding uniform sequence of ultrasonic clock intervals, accepting the binary or trinary sequence of symbolic values at the ultrasonic transducer element, and executing an encoding process at a corresponding
5 ultrasonic receiver apparatus that converts a user-specified waveform into a binary or trinary symbol sequence suitable for the transmitter to increase fidelity.

In accordance with another aspect of the present disclosure, a system is provided that includes at least one ultrasound probe configured to produce acoustic waveforms in an acoustic medium, the probes including ultrasonic transducer elements, a transmitter circuit
10 configured to accept a binary or trinary sequence of symbolic values that are configured to energize an ultrasonic transducer element with a corresponding sequence of positive, negative, or quiescent voltage levels at a corresponding uniform sequence of ultrasonic clock intervals, and a corresponding ultrasonic receiver apparatus configured to execute an encoding process configured to convert a user-specified waveform into a binary or trinary
15 symbol sequence suitable for the transmitter to achieve increased fidelity.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

The foregoing and other features and advantages of the present disclosure will be more readily appreciated as the same become better understood from the following detailed description when taken in conjunction with the accompanying drawings, wherein:

20 Figure 1 is a first embodiment of the present disclosure directed to one-way and two-way transducer compensation;

 Figure 2 is a second embodiment of the present disclosure directed to a two-way DAC synthesis usage model;

25 Figure 3 is a third embodiment of the present disclosure directed to an arbitrary waveform generation method for synthesizing stationary RF signals;

 Figure 4 is an architectural diagram of an encoding process in accordance with the present disclosure;

Figure 5 is an illustration of an example set of symbols used for an experiment in a sequel in accordance with the method of the present disclosure;

Figure 6 is an illustration of a joint IR estimate as well as individual estimates for a plurality of sequences; and

5 Figure 7 is an exemplary illustration of the method of the present disclosure implemented in a water tank.

DETAILED DESCRIPTION

In the following description, certain specific details are set forth in order to provide a thorough understanding of various disclosed embodiments. However, one skilled
10 in the relevant art will recognize that embodiments may be practiced without one or more of these specific details, or with other methods, components, materials, etc. In other instances, well-known structures or components or both associated with digital-to-analog converters and water tanks as discussed herein have not been shown or described in order to avoid unnecessarily obscuring descriptions of the embodiments.

15 Unless the context requires otherwise, throughout the specification and claims that follow, the word “comprise” and variations thereof, such as “comprises” and “comprising” are to be construed in an open inclusive sense, that is, as “including, but not limited to.” The foregoing applies equally to the words “including” and “having.”

Reference throughout this description to “one embodiment” or “an
20 embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appearance of the phrases “in one embodiment” or “in an embodiment” in various places throughout the specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any
25 suitable manner in one or more embodiments.

Transmitter Description

A brief description of the transmitter operation is given. The usage models which dictate the mathematics of the problem are described. The estimation and encoding algorithms are then introduced. The experimental approach is documented, and results are
5 then discussed.

The Vantage Ultrasound System transmitter developed by Verasonics, Inc., allows specification of arbitrary sequences of the three voltage levels $[+V, 0, -V]$ at 4 nsec clock intervals. Each acquisition event may have sequences unique to each transducer element and unique to that event. The sequences may be of arbitrary length, subject to
10 complexity of the waveform, memory limitations, and power supply capacity. A choice of internal storage formats helps economize transmitter memory usage.

A restriction in pulse sequence selection is a 3-clock minimum state dwell required to enter a positive, negative, or zero voltage level state. Another is that the achieved voltage is approximately the 5-clock running average of the voltage of the
15 achieved state.

Usage Models

The fidelity metric employed by the encoder's optimizing objective function is a design choice that depends on the usage model, or operating mode, or scenario, specific to the application. Those considered here include (1) closeness of the reference waveform
20 (in normalized RMS error) to acoustic pressure; (2) closeness of the reference waveform's predicted acoustic pressure to that actually achieved; and (3) closeness to the stationary component of the RF signal present at the input to the analog receiver gain stage. The first two problems are labeled here as the one-way transducer compensation problem, and the one-way DAC synthesis problem, respectively. Further, their two-way counterparts
25 analogously compare the reference signal to received data, rather than acoustic pressure. An illustration of the proposed invention addressing the one-way and two-way transducer compensation problem is illustrated in Figure 1. An illustration of the proposed method or process solving the two-way DAC synthesis usage model is given in Figure 2.

A third usage model uses the arbitrary waveform generation technique to synthesize stationary RF signals, considered clutter, in order to cancel them at the input to the analog receiver. This forms a differential acquisition scheme for clutter-limited applications such as Doppler imaging. This mode of operation requires additional mixing network hardware in the form of a passive low pass filter (LPF), attenuator, and summation network. Figure 3 illustrates this variation of the method. An alternative embodiment employs switchable components in the mixing network to achieve variable attenuation levels, so that the same transmitter can be used for the imaging signal path and the clutter waveform synthesis path.

10 Algorithm

The presented method of waveform encoding is motivated by the concept of symbol “equalization” when communicating through bandlimited channels [1]. This concept is generalized here to the symbol-block case. This means the entire sequence of transmit pulses (for a channel) are optimized jointly, rather than serially as individual pulses. This problem of symbol inference might be interpreted as a deconvolution constrained to discrete-valued inputs. An important difference between the communication problem and the use of the equalization concept here, is the performance metric. In the communication problem, a true symbol sequence exists against which performance can be measured (in terms of symbol error rate). In the transmitter problem addressed here, there exists no “true” symbol sequence, and the objective is only to fit the acoustic pressure generated by the transducer (or receiver data) as well as possible to the design waveform, subject to allowed inputs (symbols).

Process Architecture and Operation

The components and operation of the encoding process are shown in the architecture diagram of Figure 4. These components include IR estimation, Symbol Set definition, Symbol Set calibration, Symbol Quantization, and Equalization. The input to the process is the reference design waveform, represented by high precision samples of an

analog signal or function. The output of the process is a tri-state pulse sequence suitable for control of the transmitter pulser circuitry.

Impulse Response Estimation

Initially, the architecture requires that IR estimation be conducted for each transmit element, according to the signal path determined by the usage model. The impulse response estimation is formulated as convolution implemented in a linear statistical model and solved by least-squares theory [2]. This technique is demonstrated in prior work on underwater acoustic data at sonar frequencies [3]. Here, the model has as input a known sequence of transmitted pulses collected in a vector $\mathbf{q}$, and formed into a Toeplitz matrix [2]. Unknown parameters comprising the impulse response are represented by the vector $\mathbf{h} = [h(1), \dots, h(L)]^T$, in

$$\mathbf{Y} = \mathbf{Q} \mathbf{h} + \mathbf{e} \quad (1)$$

where the modeling error is represented by $\mathbf{e}$, the model data is the Toeplitz matrix $\mathbf{T}(\mathbf{q})$ of the zero-padded vector $\mathbf{q}$, defined as

$$\mathbf{Q} = \begin{bmatrix} q(1) & 0 & \dots & 0 \\ \vdots & q(1) & & \vdots \\ q(N) & \vdots & & \\ 0 & q(N) & \ddots & 0 \\ & 0 & \ddots & q(1) \\ \vdots & & & \vdots \\ 0 & \dots & 0 & q(N) \end{bmatrix} \quad (2)$$

and where the measurement $\mathbf{Y} = [y(1), \dots, y(N+L-1)]^T$.

One means of solving (1) is by the pseudoinverse [2], which gives the estimate of impulse response vector $\mathbf{h}$ as

$$\hat{\mathbf{h}} = \mathbf{Q}^+ \mathbf{Y} \quad (3)$$

Parameter Selection

Parameters needed for the include clipping level and symbol period, d . A design variable required by the architecture is the symbol period, which is typically based on the nominal transducer center frequency. Each symbol is comprised of several transmit
5 clock periods and therefore several tri-state voltage instances. A typical choice of symbol period corresponds to 1/4 the nominal center frequency. For example, a twelve-clock symbol period would correspond to 5.2 MHz considering a 250 MHz clock rate. Additionally, the symbol period selection determines the number of PWM levels available. The hypothetical twelve-clock symbol period permits 25 PWM levels, including the zero-
10 voltage level. The two consequences of a symbol period choice form a design tradeoff; a larger symbol period provides more PWM levels, at the cost of reduced symbol rate.

An example set of symbols, used for the experiment in the sequel, is shown in Figure 5.

The clipping level is a companion parameter to the symbol set. It adjusts
15 the overall level, with respect to the equalizer output, of the calibration gains when they are applied in the symbol quantizer. In this way the range of the equalizer output is fit within the quantizer window. The optimum clipping level is found empirically, by iteration, for each reference design waveform encoded, until the minimum mean square approximation error between the reference and synthesized waveforms is found.

20 Symbol Set Calibration

The symbol set calibration component of the architecture determines a gain mapping between each symbol available in the symbol set, and its equivalently-weighted Dirac impulse, as seen at the output of the transducer or channel model. This is achieved as the least-squares solution for the gain variable $g(k)$ in

$$25 \quad \mathbf{S}_k = g(k) \mathbf{S}_0 + \mathbf{e} \quad (4)$$

where the vector $\mathbf{S}_k$ represents the response of the impulse response model when convolved with the k -th symbol, and vector $\mathbf{S}_0$ represents the response to the prototype reference symbol, typically chosen to be the largest-bandwidth symbol.

Symbol calibration only needs to be determined once for a give symbol set and impulse response.

Equalization

The equalizer component performs a deconvolution of the desired reference design waveform against a model derived from the estimated impulse response of the usage
 5 model's signal path. The output of the equalizer comprises a sequence $\mathbf{r}$ of "soft" symbols, which represent continuously- valued Dirac impulse weights. When these are convolved with the IR model, the result approximates the specified design waveform.

In analogy to the IR estimation, a modified Toeplitz matrix formed from the
 10 estimated IR vector is used as part of a Linear Statistical Model describing convolution, so that the reference design waveform

$$\mathbf{W} = [w(1), \dots, w(L+Pd-1)]^T \quad (5)$$

is interpreted as the measurement in

$$\mathbf{W} = \mathbf{H}_d \mathbf{r} + \mathbf{e} \quad (6)$$

15 where the model matrix $\mathbf{H}_d$ is the column-decimation of the Toeplitz matrix $\mathbf{T}(\mathbf{h})$ associated with zero-padded IR response vector $\mathbf{h}$,

$$\mathbf{H}_d = \begin{bmatrix} h(1) & 0 & \dots & 0 \\ \vdots & & & \vdots \\ h(L) & \ddots & h(1) & \\ 0 & & \ddots & 0 \\ & 0 & h(L) & \ddots & h(1) \\ \vdots & & & \vdots \\ 0 & \dots & 0 & h(L) \end{bmatrix} \quad (7)$$

The decimation factor by which a subset of columns of $\mathbf{H}_d$ is retained from the standard Toeplitz matrix corresponds to the symbol period d . For example, a symbol
 20 period of 12 transmit clocks means every 12-th column of the standard Toeplitz matrix

$\mathbf{H}=\mathbf{T}(\mathbf{h})$ is retained as a column of $\mathbf{H}_d$. The unknown parameter vector (to be determined) is the collection of soft symbols in the vector $\mathbf{r}$

$$\mathbf{r} = [r(1), \dots, r(L/d+P-1)]^T. \quad (8)$$

The solution for the sequence of soft symbols in parameter vector $\mathbf{r}$ can be found by the pseudoinverse as

$$\hat{\mathbf{r}} = \mathbf{H}_d^+ \mathbf{W} \quad (9)$$

Another embodiment. For some design waveforms, an iterative extension (here labeled as the “conditional equalizer”) may give better equalization performance. In this method, the result of the equalization is quantized according to the Symbol Quantization step in the sequel. The resulting symbol sequence $\mathbf{p}$ is then applied to the impulse response in the new model notated as

$$\mathbf{W} = \mathbf{H}_d \mathbf{p} + \mathbf{e} \quad (10)$$

Then, sequentially for each element of the vector $\mathbf{p}$, the element is replaced with the symbol from the symbol set that gives the least squares residual fitting error. This is determined by exhaustive search of the symbol set, and repeated for all elements of the vector $\mathbf{p}$. If the incumbent values of the vector $\mathbf{p}$ give the lowest error, then the process stops; otherwise, it is repeated. The symbol set used in this iterative process may be of a different symbol period than that used initially, with appropriate sizing and decimation factors used.

Another embodiment. For certain reference design signals, another variant of the equalization process is used for better performance. This method, labeled here as “iterative refinement”, encodes the error signal resulting from the baseline algorithm. That is, the signal formed as the difference between the reference design signal and the achieved replica in the baseline algorithm, is treated as a new signal for synthesis by the algorithm; the motivation being that the error signal is smaller than the original signal subjected to encoding. After the error signal is encoded into a trinary sequence, its encoding is subtracted from the pulser sequence encoding of the previous stage, subject to saturation of the trinary value range. The process is terminated when the improvement in error stops.

Symbol Quantization

The symbol quantizer component chooses, for each soft symbol output sample of the equalizer, the closest symbol of the symbol set, in terms of the gain mapping determined in the calibration step. Thus, it chooses the $g(k)$ closest among each k , to the soft symbol being mapped. The sequence of symbols is then converted into their constituent trinary pulse sequences which are concatenated to a single sequence. This tri-state pulser sequence is the output of the symbol quantizer.

EXPERIMENTAL RESULTS

A water tank experiment was conducted to validate performance of the IR estimation and tri-state encoding process.

Two-way Experiment Configuration

The experiment consisted of a Philips L7-4 Transducer fixed in a water tank, and pointed directly at an acrylic block of 5.08 cm thickness, as shown in Figure 7. The Verasonics Vantage acquisition system is connected to the transducer.

Impulse Response Estimation Experiment

In the transducer impulse response estimation experiment, a transducer element channel is chosen for transmission and subsequent reception. The acrylic block acts as an acoustic mirror, reflecting substantially without modification the transmitted signal back to its source. Eight different pseudorandom pulse trains were transmitted as probing sequences, in separate acquisition events. The joint IR estimate as well as individual estimates for each sequence, were computed. These are shown in Figure 6.

LFM Synthesis Test

Waveform encoding was demonstrated on an example large time-bandwidth waveform, a Linear Frequency Modulated (LFM) pulse of 10 microsecond duration. Taylor weighting was applied to the waveform envelope. The instantaneous frequency

ranged from 3.5 MHz to 6.5 MHz. In the two-way transducer compensation usage model, the normalized RMS error between the reference waveform and the measured waveform after receiver filtering was -21.7 dB.

SUMMARY

5 In summary, a device for arbitrary waveform generation by a tristate pulser is disclosed, with application for three usage models.

 The various embodiments described above can be combined to provide further embodiments. Aspects of the embodiments can be modified, if necessary to employ concepts of the various patents, applications and publications to provide yet further
10 embodiments.

 These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the
15 full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

CLAIMS

1. A system, comprising:
 - at least one ultrasound probe configured to produce acoustic waveforms in an acoustic medium, the probes including ultrasonic transducer elements;
 - a transmitter circuit configured to accept a binary or trinary sequence of symbolic values that are configured to energize an ultrasonic transducer element with a corresponding sequence of positive, negative, or quiescent voltage levels at a corresponding uniform sequence of ultrasonic clock intervals; and
 - a corresponding ultrasonic receiver apparatus configured to execute an encoding process configured to convert a user-specified waveform into a binary or trinary symbol sequence suitable for the transmitter to achieve increased fidelity.
2. The apparatus of claim 1, wherein the encoding process is configured to accept a sampled sequence of arbitrary waveform values specified at an arbitrary numeric precision, and to provide fidelity of the specified waveform to the resulting acoustic pressure in an acoustic medium when applied to the transmitter circuit and an attached acoustic transducer probe.
3. The apparatus of claim 1, where the encoding process is configured to accept a sampled sequence of arbitrary waveform values specified at an arbitrary numeric precision, and to provide fidelity of the specified waveform to the resulting received ultrasonic signal when applied to the transmitter circuit and an attached acoustic transducer probe, and subsequently received by the ultrasonic receiver apparatus.
4. A method, comprising:
 - providing a binary or trinary sequence of symbolic values that are configured to energize an ultrasonic transducer element with a corresponding sequence of positive, negative, or quiescent voltage levels at a corresponding uniform sequence of ultrasonic clock intervals

accepting the binary or trinary sequence of symbolic values at the ultrasonic transducer element; and

executing an encoding process at a corresponding ultrasonic receiver apparatus that converts a user-specified waveform into a binary or trinary symbol sequence suitable for the transmitter to increase fidelity.

5. The method of claim 4, comprising configuring the encoding process to accept a sampled sequence of arbitrary waveform values specified at an arbitrary numeric precision, and to provide fidelity of the specified waveform to the resulting acoustic pressure in an acoustic medium when applied to the transmitter circuit and an attached acoustic transducer probe.

6. The apparatus of claim 4, comprising configuring the encoding process to accept a sampled sequence of arbitrary waveform values specified at an arbitrary numeric precision, and to provide fidelity of the specified waveform to the resulting received ultrasonic signal when applied to the transmitter circuit and an attached acoustic transducer probe, and subsequently received by the ultrasonic receiver apparatus.